PRE BOARD -1

SESSION :-2025-26

SUB:-MATHEMATICS(Code No. 041) **CLASS - XII**

Maximum Marks**: 80** Time**: 3 hours**

GENERAL INSTRUCTIONS**:**

Read the following instructions very carefully and strictly follow them:

1.This Question paper contains 38 questions. All questions are compulsory.

2.This Question paper is divided into five Sections - A, B, C, D and E.

3.In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) with only one correct option and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.

4.In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.

5.In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.

6.In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.

7.In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.

8.There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 3 questions in Section C, 2 questions in Section D and one subpart each in 2 questions of Section E.

9.Use of calculator is not allowed.

SECTION-A

This section comprises of multiple choice questions (MCQs) of 1 mark each.
Select the correct option (Question 1 - Question 18)

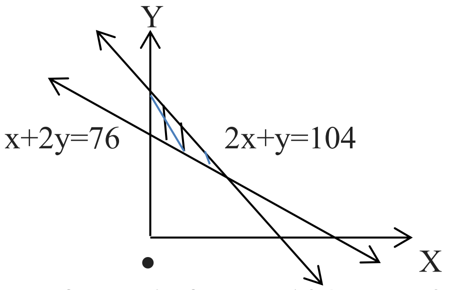
Q 1	Value of $\cos\left[\frac{\pi}{6}+\cos^{-1}\left(-\frac{1}{2}\right)\right]=\cdots$ (a) $-\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (c) $\frac{\sqrt{5}-1}{4}$ (d) $\frac{\sqrt{3}+1}{2\sqrt{2}}$	1
Q 2	If A = $\begin{pmatrix} 0 & 1 & c \\ -1 & a & -b \\ 2 & 3 & 0 \end{pmatrix}$ is a skew symmetric matrix ,then the value of a+b + c is: a) 1 b)2 c)3 d)4	1
Q 3	If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, B = $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$ then value of adj.(AB) is (a) $\begin{bmatrix} d & b \\ a & c \end{bmatrix}$ (b) $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	1
Q 4	If A is a square matrix and | A | =2 , then write the value of | AA’ |, Where A’ is the transpose of matrix A. (a) 1 (b) 4 (c) -4 (d) -1	1

Q 5	If $\begin{bmatrix} 2 & 3 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -4 & 6 \\ -9 & x \end{bmatrix}$, then the value of x is (a) 3 (b) 12 (c) 13 (d) -13	1
Q 6	f A is a square matrix of order 3 $\times$ 3 such that $\|A\| = 4$, then the value of $\|A\ (adj\ A)\|$ is (a) 4 (b) 16 (c)64 (d) 48	1
Q 7	The function $f(x) = \begin{cases} \frac{\sin x}{x} + \cos x \ , & x \neq 0 \\ k & , x = 0 \end{cases}$ is continuous at x= 0 , then the value of k is (a) 3 (b) 2 (c) 1 (d) 1.5	1
Q 8	If y = log (tanx) + log(cotx), then $\frac{dy}{dx}$ is 0 b) log(tanx) c) log(cotx) d) $\frac{1}{x}$	1
Q 9 :-	For the function y=x^3+21,the value of x, when y increases 75 times as fast as x ,is (a)±3 (b)±5√3 (c)±5 (d)none of these	1
Q 10 :-	The order and degree of the differential equation $\left(1 + 3\frac{dy}{dx}\right)^{2/3} = 4\frac{d^3y}{dx^3}$ are: (a) (1,2/3) (b)(3,1) (c) (3,3) (d) (1,2)	1
Q 11 :-	$\int_0^{\pi/2} \log(cotx)\ dx = \cdots$ (a) $\pi/4 \log tanx$ (b) $\pi/8 \log 2$ (c) 0 (d) $\pi/8 \log 8$	1
Q 12 :-	$\int \frac{e^x(1+x)}{cos^2(xe^x)} dx$ (a) $-\cot(ex^x) + c$ (b) $\tan(xe^x) + c$ (c) $\tan(e^x) + c$ (d) $\cot(e^x) + c$	1
Q 13 :-	If $\vec{a}$ and $\vec{b}$ are the position vectors of the points (1,-1,3),(-2,m,-6) find the value of m for which $\vec{a}$ and $\vec{b}$ are collinear. (a) 2 (b)1 (c) -1 (d) -2	1
Q 14 :-	The unit vector perpendicular to both the vectors $\hat{\imath} + 2\,\hat{\jmath} - 2\,\hat{k}$ and $-\hat{\imath} + 2\,\hat{\jmath} + 2\,\hat{k}$ (a) $\frac{1}{\sqrt{5}}(2\hat{\imath} - \hat{\jmath})$ (b) $\frac{1}{\sqrt{5}}(-2\hat{\imath} - \hat{\jmath})$ (c) $\frac{1}{\sqrt{5}}(2\hat{\imath} + \hat{\jmath} + \hat{k})$ (d) $\frac{1}{\sqrt{5}}(2\hat{\imath} + \hat{k})$	1
Q 15 :-	If $\|\vec{a} \times \vec{b}\|^2 = (\vec{a}.\vec{b})^2 = 400$ and $\|\vec{a}\| = 5$, then $\|\vec{b}\|$ is: (a) 3 (b) 4 (c) $4\sqrt{2}$ (d) 10	1

Q 16 :- The vertices of the feasible region determined by some linear constraints are (0, 2), (1, 1),(3, 3), (1, 5). Let Z = px+ qy where p, q> 0. The condition on p and q so that the maximum of Z occurs at both the points (3, 3) and (1, 5) is **1**

(a) p= q (b) p= 2q (c) q= 2p (d) p= 3q

Q 17:- Of the following, which group of costraints represents the feasible region given below **1**



a) X+2y≤ 76 , 2x+y≥ 104, x,y≥ 0
b) X+2y≤ 76 , 2x+y≤ 104, x,y≥ 0
c) X+2y≥ 76 , 2x+y≤ 104, x,y≥ 0
d) X+2y≥ 76 , 2x+y≥ 104, x,y≥ 0

Q 18 :- If P(A) $=\frac{3}{10}$, P(B) $=\frac{2}{5}$ and P(AUB) $=\frac{3}{5}$,then P(B/A) + P(A/B) = **1**

(a)$\frac{1}{4}$ (b) $\frac{1}{3}$ (c)$\frac{5}{12}$ (d) $\frac{7}{12}$

ASSERTION-REASON BASED QUESTIONS

(Question numbers 19 and 20 are Assertion-Reason based questions carrying 1 mark each. Two statements are given, one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the options (A), (B), (C) and (D) as given below.)

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

Q 19 :- **Assertion (A)**: $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)=\frac{\pi}{3}$ **1**

Reason (R): $\sin^{-1}(-x)=-\sin^{-1}x \ \ if \ \ x\epsilon[-1,1]$

Q 20 :- **Assertion (A):** The projection of vector $\vec{a}=\hat{\imath}+\hat{\jmath}$ on the vector $\vec{b}=\hat{\imath}-\hat{\jmath}$ is 0. **1**

Reason (R): if $\vec{a}\,.\vec{b}=0$, then the projection of vector $\vec{a}$ on the vector $\vec{b}$ is always 0.

SECTION B

This section comprises of 5 very short answer (VSA) type questions of 2 marks each.

Q 21 :-	Evaluate $tan^2(\sec^{-1} 2) + cot^2(\text{cosec}^{-1} 3)$ Or Convert the following function into simplest form $$\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right) = \cdots \quad (0 < \mathrm{x} < \frac{\pi}{2})$$	2
Q 22:-	Find the derivative of following function $\log(\frac{sinx}{1+cosx})$	2
Q 23 :-	Evaluate: $\int \frac{x\ +\ cos\,6x}{3\,x^2\ +\ sin\,6x} dx$ Or Find the area of the curve y = cos x between 0 and π.	2
Q 24:-	Find ' a ' and ' b ' , if the function given by $f(x) = \begin{cases} ax^2 + b\,,\ if\ x < 1 \\ 2x + 1\,,\ if\ x \geq 1 \end{cases}$ is differentiable at x=1	2
Q 25 :-	let $\vec{a}, \vec{b}$ **and** $\vec{c}$ be three vectors such that and $\|\vec{a}\| = 3, \|\vec{b}\| = 4$ & $\|\vec{c}\| = 5$ and each one of them being perpendicular to the sum of the other two, find $\|\vec{a} + \vec{b} + \vec{c}\|$	2
SECTION C **This section comprises of 6 short answer (SA) type questions of 3 marks each.**		
Q 26 :-	If x = a($\cos\theta$ + log tan $\frac{\theta}{2}$) and y = a sin θ, find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ OR y = x^x prove that $\frac{d^2y}{dx^2} - \frac{1}{y}(\frac{dy}{dx})^2 - \frac{y}{x} = 0$.	3 3
Q 27 :-	The radius r of a right circular cylinder is decreasing at the rate of 3cm/min and its height is increasing at the rate of 2 cm/min. When r=7cm and h=2cm find the rate of change of volume of cylinder.	3
Q 28 :-	Find the area of the region bounded by the curve $y = x^2$ and the line y=4. OR Using integration, find the area of the region bounded by the line 2y=-X+8, x-axis and the lines x=2 and x=8.	3
Q 29 :-	Find the Cartesian equations of the line passing through the point (-1, 3, -2) and perpendicular to the lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$ OR Find the foot of the perpendicular drawn from the point $2\hat{\imath} - \hat{\jmath} + 5\hat{k}$ to the line $\vec{r}$ =($11\hat{\imath} - 2\hat{\jmath} - 8\hat{k}$) + λ ($10\hat{\imath}$ -4 $\hat{\jmath}$ -11$\hat{k}$). Also find the length of perpendicular.	3

Q 30 :-	Solve the following LPP by graphical method Minimize Z = 20x +10y Subject to X+2y $\leq$ 40 3x+y $\geq$ 30 4x +3y $\geq$ 60 And x,y $\geq$ 0	3
Q 31 :-	Probability of solving specific problem independently by A and B are 1/ 2 and 1/3 respectively. If both try to solve the problem independently, find the probability that (i) the problem is solved ii) exactly one of them solves the problem.	3
SECTION D **This section comprises of 4 long answer (LA) type questions of 5 marks each**		
Q 32 :-	If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1} and using A^{-1} solve the system of linear equations x – 2y =10, 2x – y – z =8 and -2y + z =7.	5
Q 33 :-	Evaluate $\int \sqrt{\frac{\sin(x-\alpha)}{\sin(x+\alpha)}}$ OR $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{sinx+cosx}{\sqrt{sin2x}}$ dx	5
Q 34 :-	Find the particular solution of the differential equation $(x^2 + xy)\, dy = (x^2 + y^2)$ dx given that y=0 when x=1 OR $x\frac{dy}{dx} + y - x + xy\cot x = 0$	5
Q 35 :-	Find the shortest distance between the lines $\vec{r} = (4\hat{\imath} - \hat{\jmath}) + \lambda(\hat{\imath} + 2\hat{\jmath} - 3\hat{k})$ and $\vec{r} = (\hat{\imath} - \hat{\jmath} + 2\hat{k}) + \lambda(2\hat{\imath} + 4\hat{\jmath} - 5\hat{k})$	5
SECTION- E **This section comprises of 3 case-study/passage-based questions of 4 marks each with subparts. The first two case study questions have three subparts (I), (II), (III) of marks 1, 1, 2 respectively. The third case study question has 4 subparts of 1 marks each**		

Q 36 :-

Mansi and Kiran are playing Ludo at home during Covid-19. While rolling the dice, Mansi's sister Raji observed and noted that possible outcomes of the throw every time belongs to set $\{1, 2, 3, 4, 5, 6\}$. Let A be the set of players while B be the set of all possible outcomes.

$$A = \{M, K\},\ B = \{1, 2, 3, 4, 5, 6\}$$

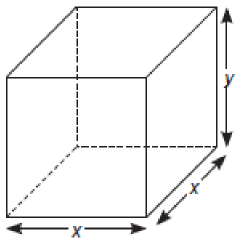
i) Let $R : B \to B$ be defined by $R = \{(x, y) : y \text{ is divisible by } x\}$.check the reflexivity of relation R. 1

ii) Check the transitivity of relation R 1

iii) Is this relation is an equivalence relation 2

OR

Let R be a relation on B defined by $R = \{(1, 2), (2, 2), (1, 3), (3, 4), (3, 1), (4, 3), (5, 5)\}$. Then check whether R is an equivalence relation. 2

Q 37 :-	A given quantity of metal sheet is to be cast in to an open tank with a square base and vertical sides as shown. Where x be the side of base square and y be the vertical side.  Answer the following based on above information: (i) If V represent the volume of tank, then find relation between V, x and y (ii) Find the total area of tank, expressed as function of x. (iii) Find x, when total volume of tank is maximum. OR If volume of tank is 1024 cm^3. The material for bottom cost at Rs5 per cm^2 and material for sides cost at Rs2.50 per cm^2, then find the value of x so that cost is minimum.	

Q38:-	A The reliability of a COVID PCR test is specified as follows: Of people having COVID, 90% of the test detects the disease but 10% goes undetected. Of people free of COVID, 99% of the test is judged COVID negative but 1% are diagnosed as showing COVID positive. From a large population of which only 0.1% have COVID, one person is selected at random, given the COVID PCR test, and the pathologist reports him/her as COVID positive.  Based on the above information, answer the following: (a) What is the probability of the 'person to be tested as COVID positive' given that 'he is actually having COVID? (b) What is the probability of the 'person to be tested as COVID positive' given that 'he is actually not having COVID'? (c) What is the probability that the 'person is actually not having COVID? (d) What is the probability that the 'person is actually having COVID given that 'he is tested as COVID positive ?	